

Titanic Survival Prediction Project

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1 Project Overview

This project aims to predict passenger survival on the Titanic disaster dataset (Kaggle). Using demographic and travel-related attributes (**Pclass**, **Sex**, **Age**, **Fare**, etc.), the task is to classify whether a passenger survived (**Survived=1**) or not (**Survived=0**).

2 How It Works

The project pipeline follows these steps:

1. Data Preprocessing:

- Missing values for **Embarked** filled with mode.
- Encoded categorical variables:
 - **Sex** mapped to numeric (male=1, female=0).
 - Extracted **title** from **Name**.
 - Extracted **cabin_num** from **Cabin**.
- Derived new features:
 - **Family_Size** = **SibSp** + **Parch** + 1
 - **Fare_per_person** = **Fare** / **Family_Size**
 - **isAlone** (1 if traveling alone, else 0)
- Encoded **Ticket** as category codes.
- Missing **Age** values predicted using a **Linear Regression model** (pipeline with scaling + **GridSearchCV**).

2. Exploratory Data Analysis (EDA):

- Pair plots of key features vs **Age**
- Correlation heatmap for numeric attributes

3. Model Training: Four different models were trained using **GridSearchCV** for hyperparameter optimization:

- Logistic Regression
- Random Forest Classifier

- XGBoost Classifier (GPU-accelerated)
 - Support Vector Classifier (SVC)
4. **Ensembling:** Final predictions are created via **weighted blending**:
- Each model's best cross-validation score is used as a weight.
 - Weighted average of predictions is computed.
 - Threshold at 0.5 decides final survival classification.

3 Technologies Used

- **Python Libraries:**
 - `pandas`, `numpy` for data processing
 - `scikit-learn` for pipelines, preprocessing, models
 - `xgboost` for gradient boosting
 - `seaborn`, `matplotlib` for visualization
- **Dataset:** Titanic dataset from Kaggle
 - `train.csv` (labeled training data)
 - `test.csv` (unlabeled test data)
 - `gender_submission.csv` (baseline sample submission)

4 Future Improvements

- Improve feature engineering (family survival groups, name-based socio-economic indicators, deck extraction).
- Use ensemble meta-learners (`StackingClassifier`) instead of weighted blending.
- Explore deep learning methods (e.g., neural networks).
- Perform Bayesian optimization for hyperparameters.